USN | | | | | | | | | | [24MTDS12]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025

DATA WAREHOUSING AND DATA MINING

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Discuss how to handle missing data, noise, and outliers in data preprocessing.	CO 1	2	**10**
	b.	Write down the multi-dimensional view of data mining.	CO 1	2	**10**
2.	**a.**	Illustrate Data Warehouse architecture and explain.	CO 2	3	**10**
	b.	Discuss the OLAP operations with an example.	CO 2	2	**10**
3.	**a.**	Illustrate the Apriori algorithm for association rule mining and discuss its limitations in terms of computational complexity.	CO 3	3	**10**
	b.	Explain the K-Nearest Neighbor algorithm with an example.	CO 3	2	**10**
		OR			
4.	**a.**	Compare the Apriori algorithm with the FP-growth algorithm. Highlight their differences in terms of efficiency and data structure used.	CO 3	3	**10**
	b.	Differentiate between random forests and decision trees.	CO 3	3	**10**
5.	**a.**	Differentiate between distance-based, and density-based approaches to anomaly detection, and explain when each technique is most effective.	CO 4	3	**10**
	b.	Explain the concept of text preprocessing in text mining. Summarize key steps such as tokenization, stemming, and stop-word removal.	CO 4	2	**10**
		OR			
6.	**a.**	Summarize the main techniques used in sentiment analysis.	CO 4	2	**10**
	b.	Analyze how sentiment analysis techniques are applied to determine public opinion on social media platforms.	CO 4	3	**10**
7.	**a.**	Explain the concept of Business Intelligence (BI) and discuss how it supports decision-making in organizations.	CO 5	2	**10**

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	b.	Describe the key components of a BI dashboard and explain its role in providing actionable insights to business users.	CO 6	2	**10**
		OR			
8.	**a.**	Summarize the role of NoSQL databases in big data integration and explain how they differ from traditional relational databases.	CO 5	2	**10**
	b.	Discuss how Hadoop is playing major role in Bigdata integration.	CO 6	2	**10**